Rank augmentation: complete measurement inputs

Human Compression project

Private input proposal; no factors

The original is Julie De Saedeleer, Dimitri Leemans and Jessica Mulpas, *A rank augmentation theorem for rank 3 string C-group representations of the symmetric groups*, arXiv:2212.12563v1, 23 December 2022. All nineteen physical pages are retained, including its title, abstract, introductory table, figures and labels, version banner, references and author addresses.

Complete papers

The scientific contract has nineteen dispositions. The first two cases of Theorem 4.1 are recovered with all their hypotheses. The third case is false as printed; the replacement includes the exact corrected criterion and the complete reviewed eighteen-point counterexample. The measured original keeps its literal statement and proof; calibration never corrects its mathematics.

The replacement pays the entire five-page common paper, the entire three-page source application and the entire three-page counterexample proof. The latter is a required written dependency, not an optional short note. No previous Topic B common theory or sibling proof is acquired free.

Complete document	Pages	Raw NFC characters
Original L020	19	33,604
Whole common paper	5	13,904
Source application	3	7,242
Whole counterexample proof	3	6,854
Whole program appendix	3	10,866

The character column counts non-whitespace characters in the untrimmed PDF extraction, the fixed primary coordinate. The five PDFs occupy 33 pages. A later written comparison must include all eleven written replacement pages. A conditional source/application comparison requires prior acquisition of the whole common and the complete declared programs; its local denominator contains both application documents, six pages in total. No factor is computed here.

Whole programs, with distinct roles

The new three-page appendix prints all three complete files once. It replaces the earlier two-page appendix; both appendices must not be added. The complete raw alternative is 11,533 bytes. Printed and raw code are alternatives. `check_examples.g` supplies required finite checks for the three Section 6 examples. `check_models.g` is bounded validation, not a premise for a theorem in unbounded degree. `check_case.py` is the transparent certificate for the fully written S_{18} obstruction. None is pruned or silently omitted because its role differs from the others.

Extraction, fonts and figures

All five established units are supplied under NFC and NFKC for untrimmed text, text with exact furniture removed, and calibrated selected text: 150 PDF cells. The selected variants remove only the isolated first source-page number, alternating repeated source headers, and isolated trailing replacement page numbers. They retain all source diagram labels and the arXiv banner. Printed hyphens and extraction spacing remain literal.

Actual PDF dictionaries, effective encodings and byte events are bound for all 528 used source glyph classes across 23 dictionaries. Twenty-four classes belong to the unembedded standard Times-Roman version banner; the remaining 504 have embedded outlines. The six actual source-space events have empty outlines; no visible whitespace glyph is discarded. Replacement Unicode Type0 text remains unchanged, without a Type1 outline claim.

Two source classes need explicit visible-symbol calibration: 111 calligraphic capital G symbols and one blackboard-bold N. Each of the 112 replacements is bound to its actual font dictionary, encoded byte, embedded outline and source occurrence. The two raw variants remain unaltered apart from declared furniture selection. This does not encode every typographic font style as mathematical Unicode. Ordinary italic variable letters remain in the established extraction representation; raw and both normalization sensitivities are preserved.

All nineteen primary pages were freshly viewed. Thirty-two exceptional and supplementary mathematical outlines, including both repaired classes, were separately rendered and read. The source has many vector diagrams; labels are extracted, but topology, colour, lines, curvature and layout are not reconstructed as text. Image/form/path presence is inventoried. Full graphics and layout costs remain unknown, not zero. No figure is removed from the numerator, and no complete-visible-graphics measurement is claimed.

Separate costs and retained limits

Three complete operational drivers total 15,947 raw bytes. Two complete Lean files total 6,855 raw bytes. Their fifteen declarations comprise two generic subgroup-factorization/intersection statements and thirteen literal output obstruction statements. They do not formalize the original input marking's C-property, generation of S_{18}, all source hypotheses or the complete theorem. These five files provide 50 separate raw metric cells; the three mathematical programs provide another 30. They are not silently added to code or paper costs.

The introductory Table 1 census remains motivating background, not a new regeneration. The source end criterion is an explicit retained classical input; unused sesquiextension background is not adopted literally. Larger middle orbits and higher ranks remain outside this recovery. Full external proofs, runtime implementations, historical software and full formalization costs are unknown. No source or question credit is added by this input packet.

The appendix has a separate distinct printing review. The input author prepared these new counts; a separate input audit is required before numerical comparison. No GAP, finite certificate or Lean computation was rerun for measurement.